



Robust Ensemble of Two Different Multimodal Approaches to Segment 3D Ischemic Stroke Segmentation Using Brain Tumor Representation Among Multiple Center Datasets

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Abstract

Ischemic stroke segmentation at an acute stage is vital in assessing the severity of patients' impairment and guiding therapeutic decision-making for reperfusion. Although many deep learning studies have shown attractive performance in medical segmentation, it is difficult to use these models trained on public data with private hospitals' datasets. Here, we demonstrate an ensemble model that employs two different multimodal approaches for generalization, a more effective way to perform on external datasets. First, after we jointly train a segmentation model on diffusion-weighted imaging (DWI) and apparent diffusion coefficient (ADC) MR modalities, the model is inferred on the DWI images. Second, a channel-wise segmentation model is trained by concatenating the DWI and ADC images as input, and then is inferred using both MR modalities. Before training with ischemic stroke data, we utilized BraTS 2021, a public brain tumor dataset, for transfer learning. An extensive ablation study evaluates which strategy learns better representations for ischemic stroke segmentation. In our study, nnU-Net well-known for robustness is selected as our baseline model. Our proposed method is evaluated on three different datasets: the Asan Medical Center (AMC) I and II, and the 2022 Ischemic Stroke Lesion Segmentation (ISLES). Our experiments are widely validated over a large, multi-center, and multi-scanner dataset with a huge amount of 846 scans. Not only stroke lesion models can benefit from transfer learning using brain tumor data, but combining the MR modalities using different training schemes also highly improves segmentation performance. The method achieved a top-1 ranking in the ongoing ISLES'22 challenge and performed particularly well on lesion-wise metrics of interest to neuroradiologists, achieving a Dice coefficient of 78.69% and a lesion-wise F1 score of 82.46%. Also, the method was relatively robust on the AMC I (Dice, 60.35%; lesion-wise F1, 68.30%) and II (Dice, 74.12%; lesion-wise F1, 67.53%) datasets in different settings. The high segmentation accuracy of our proposed method could improve radiologists' ability to detect ischemic stroke lesions in MRI images. Our model weights and inference code are available on <https://github.com/MDOpx/ISLES22-model-inference>.

Keywords Ischemic stroke segmentation · Transfer learning · Joint training · Generalization · Multimodal

Introduction

Please check the section headings if assigned to appropriate levels. Ischemic stroke is a neurological impairment that results from cerebral artery or vein occlusion [1]. The maxim “time is brain” emphasizes the importance of rapid diagnosis and treatment in these cases [2]. Early diagnosis and quantitative evaluation of stroke lesions require medical imaging [3, 4]. Magnetic resonance imaging (MRI) accurately and sensitively differentiates ischemic lesions from

stroke mimics [5], and the high sensitivity of diffusion-weighted imaging (DWI) makes it one of the most important MRI modalities for stroke evaluation [6–8]. The apparent diffusion coefficient (ADC) is a particularly useful measure because it tends to be more robust than DWI to observer variability [9, 10].

Research on deep learning-based automatic segmentation of ischemic stroke lesions has received considerable attention for its potential to reduce time and high labor costs. Although many researchers have designed DWI-based lesion segmentation models [11, 12], multimodal studies which utilize both DWI and ADC as the model's inputs have been gaining attention [13–16]. A major data source, ischemic

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stroke lesion segmentation (ISLES) challenges at the annual Medical Image Computer and Computer Assisted Intervention (MICCAI) conferences, has also been developed to promote the development of deep neural networks for identifying ischemic stroke lesions based on MR images. At the ISLES-SISS 2015 competition [17], Kamnitsas et al. [18] earned the 1st place by using 3D CNN and conditional random field (CRF) as a regularization constraint. The selection of the optimal activation functions and hyper-parameters for U-Net architecture [19] training has also been extensively studied [20]. In addition, suitable patch selection has also been found to enhance ischemic stroke lesion segmentation performance [21]. When training on an image with small infarcts using small-sized patches, class imbalance is caused by many normal patches. Therefore, sampling patches with or without stroke lesions at an appropriate rate is the key to solving the imbalance [22]. Zhan et al. [11] changed the backbone network in the U-Net architecture, and the DenseNet architecture made better results than VGG [23]. In the case of small stroke lesions, features can be lost during the encoding process, and thus an attention module has been added to U-Net for compensation [15]. In addition to 3D U-Net, 2D-based U-Net also showed desirable performance, and a combined model of 2D and 3D U-Net was also studied [24, 25]. Recently, as part of the MICCAI 2022 ISLES challenge, Ashtari et al. [14] introduced a factorizer, a model that exploits low-rank matrix factorization operations.

The Dice coefficient, a voxel-wise metric of similarity, is widely used to evaluate deep learning segmentation models. However, from a neurological point of view, using

only Dice scores to evaluate a stroke segmentation model is not preferred. For example, stroke patterns generally have unconnected areas, and lesion-wise sizes vary from small to large (Fig. 1). Thus, a high Dice score can be achieved by just segmenting a single huge lesion, even though other small lesions are missed [26]. Hence, it is essential to identify not only the number of voxels, but also the number of lesion burdens, because this information can be used as a measurement of the disease severity [27, 28]. As a result, research based on lesion-wise metrics has recently garnered interest [26, 29, 30].

In this paper, we propose an ensemble of two different multimodal methods: DWI- and ADC-based joint training and channel-wise training. For the joint training, nnU-Net [31] is jointly trained on the DWI and ADC. Before the joint training, we made pretrained model on brain tumor segmentation data (BraTS'21). For inference, only the DWI nnU-Net model was used. For channel-wise training, transfer learning with the BraTS'21 dataset was followed by training with concatenated DWI and ADC as an input data. Although Liu et al. [32] showed that using BraTS'21 data for a pre-trained model helpfully enhances the performance of single modality DWI, here we tested whether multi-channel transfer learning with DWI and ADC leads to better.

Our ensemble model was evaluated on one public and two private datasets: the Ischemic Stroke Lesion Segmentation 2022 challenge (ISLES'22) dataset [26], and the Asan Medical Center (AMC) I and II. Applying brain extraction tool (BET) in a clinical setting has been limited in hospitals because it can unintentionally eliminate important brain

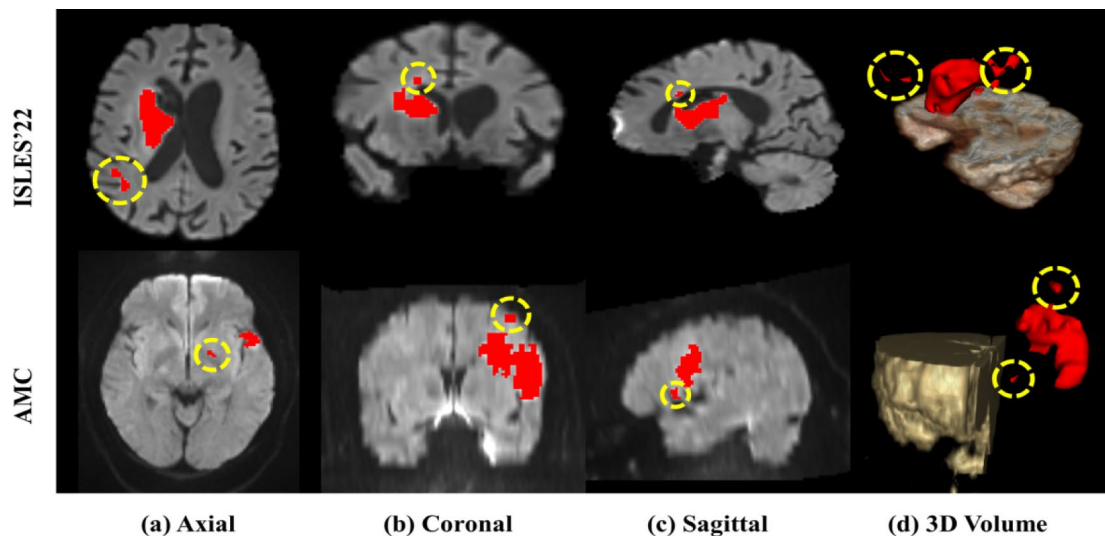


Fig. 1 Three DWI planes and two 3D volumes are visualized to show ischemic stroke lesions. BET (brain extraction tool) is applied on only the ISLES'22 data (ischemic stroke lesion segmentation) to remove noises and artifacts. Both datasets have similar stroke lesion

variations, including large territorial infarcts and small punctiform ischemic lesions. Small lesions are indicated with yellow dotted circles

lesions and is time-consuming. Therefore, for external validation, we use realistic clinical images without the BET. Four metrics, including the Dice score, lesion count difference, lesion-wise F1 score, and volume difference, are selected to comprehensively evaluate our models.

Materials and Methods

Datasets

Four different data sources were used: BraTS'21 [33] for pre-training, ISLES'22 [26] for internal validation, and AMC I and AMC II for external. The AMC I and II datasets were collected according to the principles of the Declaration of Helsinki, and the data collection was performed in accordance with current scientific guidelines. The study protocol was approved by the Institutional Review Board (IRB) Committee of AMC, University of Ulsan College of Medicine, Seoul, Republic of Korea. The requirement for informed patient consent was waived by the IRB Committee of AMC [34].

BraTS'21

The BraTS'21 data includes 2040 multiparametric MRI (mpMRI) scans separated into 1251 training, 219 validation, and 570 test datasets. Because the ground truths of the validation and test data are not open to the public and one training data has an issue, for pretraining experiments we utilized the 1250 training data, split into 1000 training and 250 validation sets. The data contains four modalities for each scan: T1 naïve (T1), T2-weighted (T2), T2-FLAIR, and post-contrast-T1-weighted (T1-CE). The data consists of three classes: [1] non-enhancing and necrotic tumor core (necrosis), [2] peritumoral edema (edema), and [3] Gd-enhancing tumor (enhancing tumor). After registration to the Montreal Neurological Institute (MNI) template [35], the mpMRI volumes were resampled to be 1 mm³ with an isotropic resolution and skull-stripped. To begin manual annotations,

automated segmentation volumes on four mpMRI scans were given to neuroradiologists who initially refined the volume. Two senior neuroradiologists assessed the automated segmentations for further refinements.

ISLES'22 The ISLES'22 challenge offered ischemic stroke data from three medical centers: University Hospital of the Technical University of Munich (TUM), University Hospital of Bern (Bern), and University Medical Center Hamburg-Eppendorf (UKE). TUM and UKE provided acute to sub-acute stroke MRIs, all acquired after revascularization therapy, and Bern provided hyper-acute to acute stroke MRIs from before revascularization. The ISLES'22 data consists of 400 mpMRI scans separated into 250 training data and 150 test data [26]. The training data was from TUM and Bern, while the test data was from all three centers. Note that we used primarily the DWI and ADC because the image registrations between the FLAIR and DWI were complicated and inaccurate.

AMC I and II

The AMC data was used to demonstrate our model's performance on real hospital data. Our external data included mostly South Korean patients. All the MR parameters are shown in Table 1.

AMC I: For external validation, a dataset of brain DWI and ADC MR images from 267 patients with ischemic stroke was acquired between January 2019 and December 2021. An investigator manually segmented the DWI high-signal intensity area thought to be ischemic lesion, using the MRIcron image viewer [36]. Afterwards, a neurologist revised the ischemic lesion mask. Each of the investigators had more than 10 years of experience with stroke imaging.

AMC II: Between September 2005 and August 2015, a dataset of brain DWI and ADC MR images was collected from 429 ischemic stroke patients. Two neuroradiologists, with 6 and 9 years of experience, respectively, agreed on the definition of the acute infarct lesions shown on the DWI. To

Table 1 Overview of MR imaging parameters from five centers for ischemic stroke lesion segmentation evaluation. The parameter values are shown in [min–max]

Center	TR (ms)	TE (ms)	Flip angle (°)	Voxel (mm ²)	Slice thickness (ms)	Gap (mm)
TUM	[3175–16439]	[55–91]	[90–90]	[0.87×0.87–2.00×2.00]	[2.00–5.00]	[2.00–5.00]
Bern	[3500–4780]	[67–80]	[90–180]	[0.57×0.57–2.00×2.00]	[3.00–6.50]	[4.80–4.80]
UKE	[13000–13000]	[90–90]	[90–90]	[0.89×0.89–1.87×1.87]	[2.00–6.50]	[0.00–0.00]
AMC I	[3000–9690]	[56–91]	[90–180]	[0.65×0.65–1.30×1.30]	[3.00–5.00]	[0.00–2.00]
AMC II	[400–8000]	[16.1–103]	[18–90]	[0.45×0.45–1.30×1.30]	[3.00–5.00]	[0.00–2.50]

We have three different cohorts: ISLES'22 (TUM, Bern, UKE), AMC I, and AMC II. *TUM* University Hospital of the Technical University Munich (Germany), *Bern* University of Hospital of Bern (Switzerland), *UKE* University Medical Center Hamburg-Eppendorf (Germany), *AMC I* Department of Neurology, Asan Medical Center (South Korea), *AMC II* Department of Radiology, Asan Medical Center (South Korea)

exclude areas of T2 shine-through on the DWI, the neuroradiologists defined the maximum visual extent of high-signal intensities on b1000 images as an infarct lesion [35].

Preprocessing

First, we cropped the foreground portion that corresponded to the brain region. To keep values outside a region of interest (ROI) from affecting normalization, the scope of the normalization was limited to the ROI, a process called non-zero mask normalization. Z-score normalization was used to handle outliers well by making the standard close to zero and the standard deviation 1. Finally, to prevent overfitting, various augmentations were applied: rotation, elastic deformation, horizontal flip, scaling, random cropping, Gaussian blur, Gaussian noise, brightness, contrast, and random gamma transform.

nnU-Net

Due to automatic configuration setting as preprocessing and hyper-parameters for model architecture, nnU-Net has gained substantial tractions as a medical 3D segmentation model for developers with limited experience in medical imaging [31]. nnU-Net has the ability to automatically adapt and configure itself for various tasks, eliminating need for extensive domain knowledge. nnU-Net's generalization performance has been extensively validated across diverse datasets and is actively used for various medical image segmentation tasks [37–39]. Although neural architecture search (NAS), transformer, residual block, and ConvTransformer-based models are well-known for the state-of-the-art (SOTA) 3D medical segmentation models [40–46], it usually requires a large dataset and training

time. In our study, nnU-Net methods show better results than others as shown in Table 2, so we selected nnU-Net as our baseline model and all experiments are conducted by nnU-Net.

Figure 2 shows an overview of a nnU-Net. The loss function includes combination of binary-cross entropy and soft dice loss. The cross-entropy loss is defined as follows:

$$L_{bce}(y, \hat{y}) = -(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})) \quad (1)$$

where y notes the ground truth label and $\hat{y}$ the prediction probability. Here, the ground truth is the ischemic stroke lesion mask and has a value of 0 or 1. The prediction probability is the probability predicted by the label and has a continuous value between 0 and 1. Soft dice loss is also defined by ground truth y , prediction probability $\hat{y}$, and number of voxels N :

$$L_{dice}(y, \hat{y}) = 1 - \frac{2 \sum_{n=1}^N y_n \hat{y}_n}{\sum_{n=1}^N y_n + \sum_{n=1}^N \hat{y}_n} \quad (2)$$

Additionally, the nnU-Net adopts deep supervision (DS) concept to utilize multi-scale features with skip-connections. Coefficients for DS are determined by the formula below:

$$a_i = (2^{i-1} \times \sum_{j=1}^{N-1} 2^{j-1})^{-1} \quad (3)$$

$$DS = \alpha_i \sum_{i=1}^{N-1} (L_{bce} + L_{dice}) \quad (4)$$

where N is the number of down-sample layers. This means that an exponentially decreasing weight is assigned to each output as the resolution decreases.

Table 2 Comparison with 3D medical imaging segmentation models on the ISLES'22 test data for single and multi-channel training. $\oplus$ represents the concatenation operation. The best results are shown in bold, and the second-best are underlined. Using the Mann–

Whitney–Wilcoxon test, statistical significance is indicated by a p -value calculated between nnU-Net vs. others. (*: p -value < 0.05; **: p -value < 0.01; ***: p -value < 0.001; and ****: p -value < 0.0001)

Model	Dice ($\uparrow$)		Lesion-wise F1 ($\uparrow$)		Lesion count difference ($\downarrow$)		Volume difference ($\downarrow$)	
	DWI	DWI $\oplus$ ADC	DWI	DWI $\oplus$ ADC	DWI	DWI $\oplus$ ADC	DWI	DWI $\oplus$ ADC
ResU-Net [40]	62.07****	63.37****	43.71****	<u>43.98****</u>	6.96****	6.68****	<u>6.48*</u>	<u>6.80***</u>
STU-Net [41]	<u>67.01****</u>	66.88****	39.31****	38.87****	7.66****	7.65****	7.67****	6.9***
DiNTS [42]	59.58****	61.93****	<u>44.01****</u>	43.65****	12.20****	8.82****	11.30****	9.91****
UNETR [43]	58.25****	58.87****	34.68****	34.06****	19.84****	18.75****	10.53****	9.41****
Swin UNETR [44]	64.83****	<u>67.12****</u>	38.01****	37.13****	22.42****	22.38****	8.25***	7.33****
UTNet [45]	62.93****	55.77****	43.84****	30.32****	12.28****	51.37****	8.80****	45.30****
nnFormer [46]	47.55****	53.84****	38.82****	41.81****	<u>6.04****</u>	<u>5.96****</u>	13.35****	9.67****
nnU-Net [31]	76.41	77.18	77.51	77.45	2.52	2.18	5.46	4.80

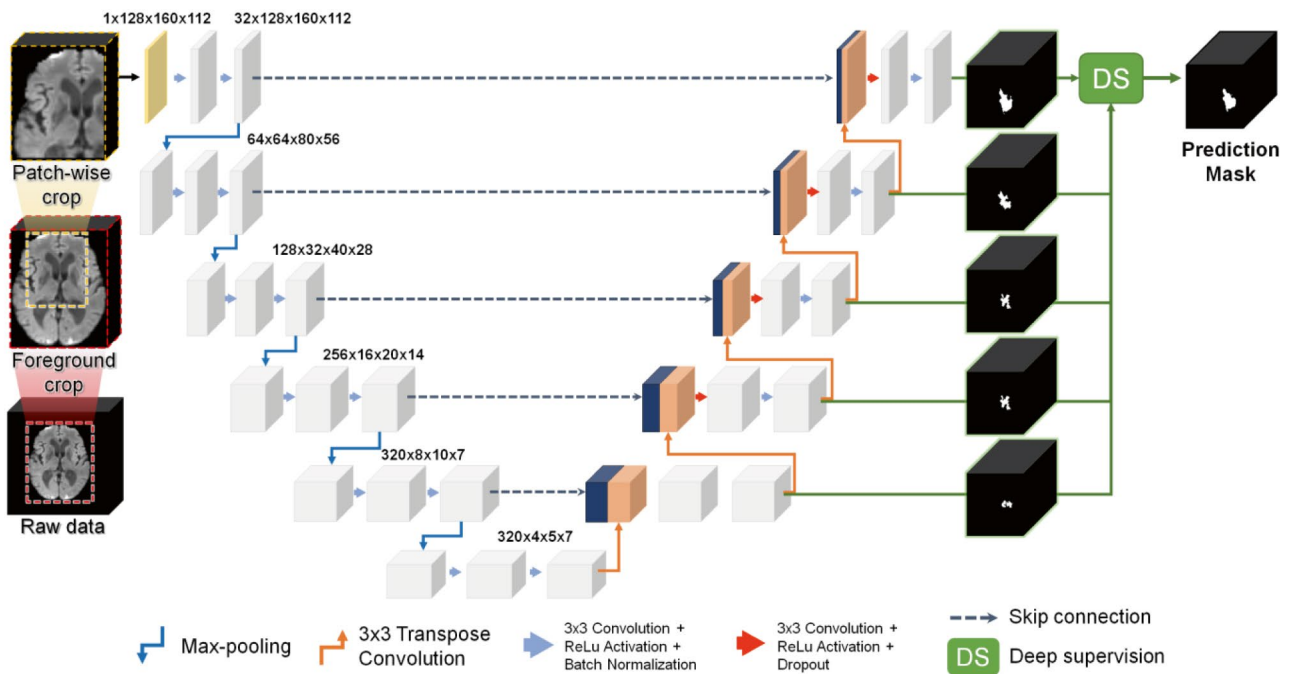


Fig. 2 Overview of a deep supervised network. Based on GPU VRAM specifications, the network and patch size are automatically determined. To train multi-scale resolution, deep supervision is applied with coefficients. All processes and results are based on 3D volume

The Proposed Methodology

Knowledge Transfer Based on Brain Tumor

The BraTS'21 data includes various MRI modalities and brain tumor types. Assuming the brain tumor data as the source domain (D_S) and brain tumor segmentation as the source task (T_S), the data from ISLES'22 and stroke lesion segmentation task is denoted as the target domain (D_T) and the target task (T_T). Transfer learning allows the target model to learn general knowledge from D_S and T_S . The BraTS'21 dataset contains four MRI modalities (e.g., T2 fluid-attenuated inversion recovery (FLAIR), T1

contrast-enhanced, T1 naïve, and T2-weighted images) and three different classes (e.g., necrosis, peritumoral edema, and enhancing tumor). Before applying channel-wise training for multimodal inference (ADC and DWI) or joint training for single modal inference (DWI), transfer learning is performed for the target task. We conducted transfer learning for an ablation study to find an optimal pretrained model for ischemic stroke lesion segmentation. We evaluated each of the pretrained models, using the ISLES'22 test dataset to determine which pretrained model gave the best results. Tables 3 and 4 give details of two ablation cases: transfer learning for joint training and multi-channel. To assess the performance of the pretrained model, we used

Table 3 Transfer learning ablation study on the ISLES'22 test data to find the optimal modality. In the ablation study, conducted on the necrosis class of the BraTS'21. When we applied transfer learning for multi-channel models, the same modality as source data is concatenated. The best score of joint and multi-channel model is respectively

Source Data	Dice (↑)		Lesion-wise F1 (↑)		Lesion count difference (↑)		Volume difference (↑)	
	Joint	Multi-channel	Joint	Multi-channel	Joint	Multi-channel	Joint	Multi-channel
w/o transfer	76.41**	77.18***	77.51*	77.45**	2.52	2.70***	5.46	4.944
T1 Naïve	77.69	77.29	80.35	80.42	2.30	2.17	5.82	4.809
T2 FLAIR	77.87	77.32	80.21	79.83	2.10	2.16	5.58	5.307
T1 CE	77.74	76.56*	79.05	78.83	2.24	2.24	5.55	4.980
T2	77.51	77.78	79.29	80.50	2.18	2.18	5.70	4.809

shown in red and blue bold. Using the Mann–Whitney–Wilcoxon test, statistical significance is indicated by a p -value calculated between T2 FLAIR vs. others in joint, and T2 vs. others in multi-channel. (*: p -value < 0.05; **: p -value < 0.01; ***: p -value < 0.001; and ****: p -value < 0.0001)

Table 4 Results of a transfer learning ablation study on the ISLES'22 test data, seeking the best class. The best score of joint and multi-channel model is respectively shown in red and blue bold. Based on Table 3, we respectively selected T2 FLAIR and T2 as an opti-

Class	Dice ($\uparrow$)		Lesion-wise F1 ($\uparrow$)		Lesion count difference ($\uparrow$)		Volume difference ($\uparrow$)	
	T2 FLAIR	T2 $\oplus$ T2	T2 FLAIR	T2 $\oplus$ T2	T2 FLAIR	T2 $\oplus$ T2	T2 FLAIR	T2 $\oplus$ T2
Necrosis	77.87	77.78	80.21	80.50	2.10	2.18	5.58	4.80
Edema	78.29	76.82**	79.70	79.15*	2.28	2.30	5.55	4.96
Enhancing tumor	77.24	76.93*	79.41	80.46	2.20	2.19	5.40	5.30

the DWI data for the joint training experiment and the DWI and ADC data for the multi-channel experiment.

Two Different Multimodal Approaches: Joint Training and Channel-Wise Training

We designed two different multimodal approaches, the joint training and channel-wise training. For the joint training, the nnU-Net receives only a single modality, while multimodal inputs, the concatenation of DWI and ADC, are needed in the channel-wise training.

Joint training: Here, we suggest a new stroke multimodal approach by simultaneously training the DWI and ADC models. We utilize both the DWI and ADC in the training phase, but only use the DWI in the inference phase (Fig. 3(a)). This is because we postulate that taking both modalities for external datasets in the inference phase could make generalizability worse because the model considers the variances of both DWI and ADC. The joint training loss is defined as follows:

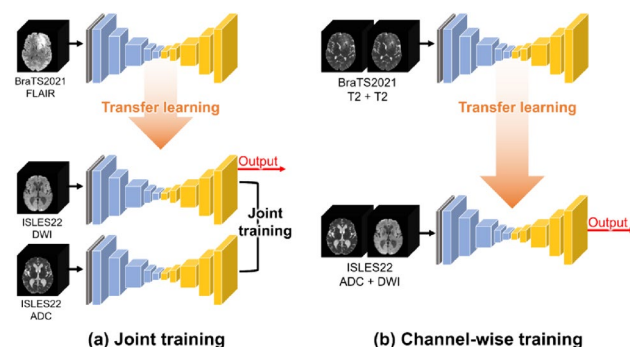


Fig. 3 (a) An overview of joint training. After selecting the optimal pretrained model on brain tumor, we simultaneously applied transfer learning and joint training with two MRI modalities, DWI and ADC. Finally, DWI is only considered an inference modality. (b) The overview of the channel-wise training scheme. The pretrained model is trained using two same modalities from BraTS'21 for transfer learning, and then the model is fine-tuned by taking as input multimodal data that is a concatenation of DWI and ADC. Black arrows point to the training phase, while red arrows show inference-phase operation

mal modality of joint training and multi-channel. Using the Mann-Whitney-Wilcoxon test, statistical significance is indicated by a p-value calculated between Necrosis vs. others. (*: p-value < 0.05; **: p-value < 0.01; ***: p-value < 0.001; and ****: p-value < 0.0001)

$$L_{joint\ training} = \frac{1}{2} \sum_{i=1}^N (DS_{DWI}^{(i)} + DS_{ADC}^{(i)}) \quad (5)$$

where N is the batch size.

Channel-wise training: The channel-wise training simply concatenates different MRI modalities for training (Fig. 3(b)). Both the DWI and ADC are used for the inference step as opposed to joint training, but all other environments are the same as those in the joint training except that the input size is changed to $128 \times 160 \times 112 \times 2$ through a pre-processing stage.

Experiment Environment and Training Setup

All models were implemented using Pytorch, and experiments were performed on an NVIDIA A100 GPU with 80 GB. The patch size for input was set to $128 \times 160 \times 112 \times M$, where M is the number of MRI modalities. We trained all models for 1000 epochs, with a batch size of 2. Before the transfer learning, the network was initialized by the method of Kaiming He. Adam optimizer was selected with a learning rate of 0.01, a weight decay of $3e-05$, and a momentum of 0.99. The learning rate was decreased during training, which followed the poly learning rate (PolyLR) schedule with an exponent of 0.9 [47]. The setting of these parameters is followed by default in nnU-Net.

$$Learning\ rate = initial\ learning\ rate \times (1 - \frac{epoch}{max_{epoch}})^{exponent} \quad (6)$$

Evaluation Metrics

To evaluate our model's performance, we used voxel-wise and lesion-wise metrics. The voxel-wise metrics included the Dice score and the volume difference. The volume difference is the absolute difference between the total number of predicted voxels and the ground truth voxels. The lesion-wise metrics, lesion-wise F1 score, and lesion count difference were used to evaluate how well the model found unconnected lesions. The lesion count difference measures the absolute difference between the total number of predicted unconnected lesions and that of the ground truth.

Results

Finding the Best Pretrained Model on BraTS'21

The quantitative results of the two different transfer learning experiments are shown in Table 3 and 4.

As presented in Table 3, the best choice for the pretraining step before joint training is T2 FLAIR among different MR modalities. Although Table 3 shows slight differences between the four modalities' outcomes, we confirm that transfer learning is generally successful as measured by three metrics: Dice, lesion-wise F1, and lesion count difference. In addition, T2 is the optimal MR modality for multi-channel training by enhancing the Dice, lesion-wise F1, lesion count difference, and volume difference.

Segmentation Results for ISLES'22

Table 2 shows all results of SOTA models trained from scratch. Except for UTNet and STU-Net, using DWI and ADC gives better Dice score than only DWI. nnU-Net model not only has a better Dice score than the other models, but also outperforms the existing SOTA models, especially lesion wise, showing that it better locates the lesions of interest and does not predict unnecessary lesions.

In the top section of Table 5, the Mann–Whitney–Wilcoxon tests are used to statistically compare the other models with “(g) Ensemble,” the ensemble model with joint training (d) and channel-wise training (f). Transfer learning strategies (b) and (d) are superior to scratch-trained models, including (a) and (c), in three metrics: Dice, lesion-wise F1,

Table 5 Quantitative segmentation results from three different datasets and ablation studies. The best results are shown in bold, and the second-best are underlined. (g) represents an ensemble of (d) and (f). Using the Mann–Whitney–Wilcoxon test, statistical significance is indicated by a *p*-value calculated between (g) vs. others. (*: *p*-value < 0.05; **: *p*-value < 0.01; ***: *p*-value < 0.001; and ****: *p*-value < 0.0001)

	Dice (↑)	Lesion-wise F1 (↑)	Lesion count difference (↓)	Volume difference (↓)
ISLES'22 Challenge				
(a) DWI (1)	76.41****	77.51***	2.52	5.46
(b) DWI (2)	77.87****	80.21**	2.10	5.58*
(c) DWI ⊕ ADC (3)	77.18****	77.45****	2.70*	<u>4.94</u>
(d) DWI ⊕ ADC (4)	77.78*	80.50**	2.18	4.80
(e) DWI (5)	76.47****	77.28****	2.52	5.70*
(f) DWI (6)	<u>78.25</u> ***	<u>81.61</u>	<u>2.14</u>	5.54
(g) Ensemble	78.69	82.46	2.28	5.10
AMC I				
(a) DWI (1)	39.79 ****	48.14 ****	<u>2.19</u>	1554.91 *
(b) DWI (2)	42.25 ****	43.41 ****	5.57****	1998.92 ****
(c) DWI ⊕ ADC (3)	39.24 ****	43.08 ****	2.84***	1206.24 *
(d) DWI ⊕ ADC (4)	51.15 **	50.86 ****	2.92****	1255.39
(e) DWI (5)	34.66 ****	39.34 ****	2.38	1519.16 **
(f) DWI (6)	<u>59.24</u>	<u>63.17</u>	2.52 **	767.40
(g) Ensemble	60.35	68.30	1.78	<u>806.76</u>
AMC II				
(a) DWI (1)	58.22 ****	57.74 ****	3.48	2390.69
(b) DWI (2)	46.63 ****	41.38 ****	9.92****	11,832.39****
(c) DWI ⊕ ADC (3)	29.35 ****	30.84 ****	5.57****	5142.01****
(d) DWI ⊕ ADC (4)	64.30 **	<u>60.26</u> ****	<u>3.45</u>	2900.54*
(e) DWI (5)	54.03 ****	55.70 ****	3.60	2810.25**
(f) DWI (6)	<u>72.70</u>	50.21 ****	6.91****	<u>1765.40</u> **
(g) Ensemble	74.12	67.53	2.91	1564.20

(a) DWI [1] is the segmentation result used to evaluate the model trained with the DWI modality of ISLES'22 using the basic nnU-Net on test data. (b) DWI [2] is the result for transfer learning from T2 FLAIR of the BraTS'21 to ISLES'22 DWI. (c) DWI ⊕ ADC [3] is the result of channel-wise training. Also, (d) DWI ⊕ ADC [4] is the result of transfer learning from two channel-wise T2 of the BraTS'21 to ISLES'22 DWI and ADC. (e) DWI [5] is the result of joint training. (f) DWI⁶ is the result of joint training with transfer learning from T2 FLAIR to DWI and ADC. (g) Ensemble is the result of the ensemble of the two multimodal methods, (d) and (f)

Table 6 Top 10 teams on ISLES 2022 challenge. The average Dice, lesion-wise F1, lesion count difference, and volume difference are reported for every team. The ranking is calculated for each metric, and all metric rankings are averaged (Avg) to determine the final ranking. The best score for each metric is shown in bold

	Rank (Avg)	Team	Dice (Rank)	Lesion-wise F1 (Rank)	Lesion count difference (Rank)	Volume difference (Rank)
1	(1.75)	PAT (Ours)	78.69 [1]	82.46 (1)	2.28 (1)	5.10 (4)
2	(2.25)	SEALS	78.06 (3)	80.42 (2)	2.32 (2)	4.97 (2)
3	(2.75)	NAUTO (48)	78.39 (2)	78.82 (3)	2.54 (3)	5.04 (3)
4	(4.25)	Swan (14)	76.33 (4)	75.26 (4)	2.88 (4)	5.14 (5)
5	(5.5)	PLORAS	71.24 (7)	69.32 (6)	3.76 (8)	4.85 (1)
6	(6.25)	VICOROB	72.25 (6)	69.35 (7)	3.54 (6)	5.52 (6)
7	(6.25)	CTRL-KCL	74.28 (5)	74.16 (5)	3.04 (5)	64.69 (10)
8	(7.5)	MIRC	70.35 (8)	68.51 (8)	3.66 (7)	7.13 (7)
9	(9)	CREATIS	68.82 (9)	67.17 (9)	4.40 (9)	7.86 (9)
10	(9.5)	Dolphins	67.90 (10)	57.35 (10)	5.51 (10)	7.27 (8)

and lesion count difference. The transfer learning strategy especially improves the lesion-wise F1, by about 3%. The joint training–only strategy (e) has a lower lesion-wise F1 value than the scratch-trained model (a), and poorer lesion-wise metric scores than the transferred model (b). On the other hand, (f), with both transfer learning and joint training applied simultaneously, outperforms the model with only transfer learning (b), except for the lesion difference count. Compared to transfer learning with multi-channels (d), transfer learning with joint training (f) yields more favorable results in terms of the Dice, lesion-wise F1, and lesion count difference. Ensemble (g) of the two different multimodal methods shows the best Dice and lesion-wise F1 scores of all. In particular, the ensemble (g), which includes the strengths of (d) and (f), shows statistically significant differences from model (d) (p -value < 0.05) and model (f) (p -value < 0.001). Notably, the ensemble (g) was submitted to the ISLES'22 challenge and earned the number 1 ranking, as determined by the mean rank across all metrics

(Table 6). The qualitative results are shown in Fig. 4. Our proposed model is ranking first in the ISLES'22 ongoing challenge (leaderboard accessed on 19.07.2023, <https://isles22.grand-challenge.org/>, team-name PAT), showing good generalization to ISLES external datasets.

Segmentation Results for AMC I and II

The middle and bottom sections of Table 5 show that the ensemble (g) of two different multimodal methods is most robust among all other methods for the AMC I and II data, with respect to background noises and artifacts, respectively.

The quantitative results of the AMC I data are detailed in the middle section of Table 5. Compared to the results from the ISLES'22 challenge, the scratch-trained models, including (a) and (c), do not perform well on the AMC I data. When it comes to the Dice metric, all the transfer learning strategies are effective, but in terms of the lesion-wise F1 metric, the results are improved only when transfer learning

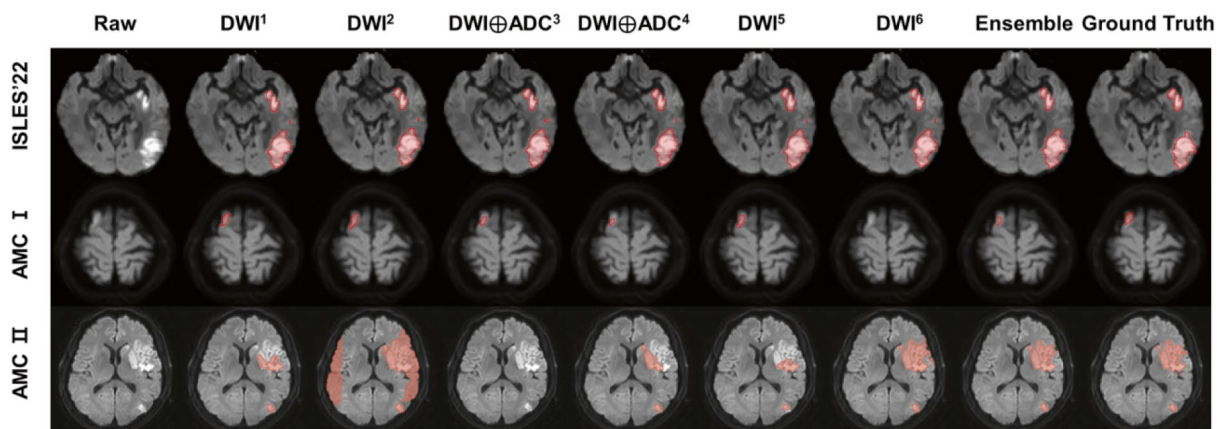


Fig. 4 Qualitative results of ischemic stroke lesion segmentation on ISLES'22, AMC I, and AMC II. The segmentation results from the ensemble method are most similar to the ground truth

is applied to multi-channel training (d) and joint training (f). Transfer learning with joint training (f) performs better on the Dice and lesion-wise F1 metrics than transfer learning with the multi-channel (d), by 8% and 13%, respectively. The ensemble (g) of (f) and (d) shows statistically significant difference only with (d) on the Dice (p -value < 0.01) and on the lesion-wise F1 (p -value < 0.0001).

The quantitative results for the AMC II data are detailed in the bottom section of Table 5. DWI-based transfer learning (b) performs worse than scratch training (a), while multi-channel transfer learning (d) yields dramatically enhanced performance. The Dice score of transfer learning with joint training (f) shows that the effectiveness of joint training can be maximized, depending on which pretrained model is used. Similar to the AMC I data, the ensemble (g) achieves the best performance, with statistically significant difference in the Dice score compared with (d) (p -value < 0.01). On the lesion-wise F1, the ensemble model (g) also demonstrates a statistically significant difference compared with all the other methods (p -value < 0.0001).

We also compared our ensemble method with recent ensemble methods for a brain segmentation task. Uncertainty ensemble [50] and multi-model ensemble [51] require a huge number of parameters, especially showing poor results at lesion-wise metrics and volume-wise (Table 7). LST-AI [53] with three 3D-UNets and Self-ensemble [52] from two different training pipelines have a greatly smaller model size than two previous ones, but still performing significantly worse than our ensemble across all metrics. Despite the fact that ensemble models typically require a lot of computation, our model with DWI and ADC modality shows not only good performance but also low computation,

reducing hardware requirements and thus saving costs. Furthermore, the computation increases proportionally with the input data size, demonstrating how well our model can compute on large and complex datasets.

To analyze our ensemble method, two ablation studies are conducted. We examined which model should have a higher weight when ensembling a jointly trained model and a channel-wise model. Table 8 shows that the ensemble result is gradually worse than the jointly trained model (Table 5(f)) in the AMC I data as the weight of the channel-wise model increases. In AMC2, we see that while the lesion-wise metric results get declined with increasing weight of the jointly trained model, the volume metric results improve. These results confirm that when ensembling the two different multimodal models, it is desirable to weight them equally. Additionally, when jointly training the ADC model and DWI model, we conducted an ablation experiment to find the desirable weights for both models. Table 9 also demonstrates that the performance is generally better when jointly learning the ADC model than when transfer learning using only the DWI model.

Discussion

Accurate segmentation of ischemic stroke images can aid diagnosis of stroke and treatment outcomes. Automated segmentation with deep learning can alleviate high labor costs, but insufficient training data and low generalization performance make it difficult to be practically applied in the clinical setting. In this study, we present an ensemble method that surmounts these difficulties by leveraging two different representations—from

Table 7 Comparison with recent ensemble strategies from two external datasets. Except for our ensemble, all ensembles only utilize a concatenation of DWI and ADC as input data. The best results are shown in bold, and the second-best are underlined. Using the Mann–

Whitney–Wilcoxon test, statistical significance is indicated by a p -value calculated between ours vs. others. (*: p -value < 0.05; **: p -value < 0.01; ***: p -value < 0.001; and ****: p -value < 0.0001)

Model	Dice (↑)	Lesion-wise F1 (↑)	Lesion count difference (↓)	Volume difference (↓)	Parameter
AMC I					
Uncertainty ensemble (50)	40.85****	33.25****	5.88****	1270.28****	296 M
Multi-model ensemble (51)	36.13****	42.76****	4.14****	2029.52****	267 M
Self-ensemble (52)	<u>50.58</u> ****	54.10****	3.25****	1694.87**	69 M
LST-AI ensemble (53)	46.32****	<u>61.90</u>	<u>2.67</u>	1956.62**	73 M
Ours	60.35	68.30	1.78	806.76	61 M
AMC II					
Uncertainty ensemble (50)	23.42****	22.36****	5.95****	5990.93****	296 M
Multi-model ensemble (51)	30.36****	23.97****	6.94****	5451.17****	267 M
Self-ensemble (52)	<u>67.30</u> ****	<u>51.03</u> ****	<u>5.20</u> ****	<u>2248.87</u> ***	69 M
LST-AI ensemble (53)	63.31****	47.11****	5.50****	2643.89*	73 M
Ours	74.12	67.53	2.91	1564.20	61 M

Table 8 Ensemble ablation study. Our final ensemble model is the average ensemble of the joint (Table 5(f)) and channel-wise model (Table 5(d)). Considering a result probability of the channel-wise model as Model A and the joint model as Model B, respectively, the final ensemble probability is derived by expression $0.5 * \text{Model A} + 0.5 * \text{Model B}$

Model A / Model B weight	Dice ($\uparrow$)	Lesion-wise F1 ($\uparrow$)	Lesion count difference ($\downarrow$)	Volume difference ($\downarrow$)
AMC I				
0.1 / 0.9	59.40	64.50	2.46***	766.13
0.2 / 0.8	59.68	66.14	2.19	765.32
0.3 / 0.7	60.01	66.59	2.10	<u>765.76</u>
0.4 / 0.6	<u>60.13</u>	<u>66.59</u>	<u>2.05</u>	771.68
0.5 / 0.5 (Ours)	60.35	68.38	1.78	806.76
0.6 / 0.4	56.89	61.02**	2.24**	974.80
0.7 / 0.3	54.51	57.15****	2.50****	1066.66
0.8 / 0.2	53.24	54.50****	2.64****	1139.52
0.9 / 0.1	52.11	52.63****	2.75****	1200.93
AMC II				
0.1 / 0.9	73.49*	52.02****	6.29****	1680.13**
0.2 / 0.8	74.22	54.97****	5.50****	1609.18
0.3 / 0.7	<u>74.79</u>	58.16****	4.78****	<u>1549.03</u>
0.4 / 0.6	75.11	61.57***	4.13****	1502.39
0.5 / 0.5 (Ours)	74.12	67.53	2.91	1564.20
0.6 / 0.4	70.43**	<u>65.15</u>	<u>2.93</u>	2068.74
0.7 / 0.3	68.54****	63.59*	3.13	2348.53
0.8 / 0.2	66.93****	62.46**	3.24	2573.22*
0.9 / 0.1	65.52****	61.40***	3.33	2749.19*

joint training and channel-wise training—with the knowledge gained from brain tumor segmentation.

Searching optimal parameter settings is practically important to apply a model on clinical domain since it greatly affects performance. However, conducting a huge number of combinations for the optimal setting requires time-consuming works. nnU-Net automatically finds optimal

parameter settings such as the size of patch and network, showing a good performance at 11 international medical segmentation challenges. Therefore, we empirically set the parameter settings for our single (DWI) and multimodal (DWI $\oplus$ ADC) model, depending on GPU memory. Table 2 demonstrates that nnU-Net is considerably robust to external dataset. While transformer-based models (e.g. UNETR,

Table 9 Joint training ablation study. Based on Eq. (5), we experimented with the joint training by setting different weights for the ADC and DWI models. DWI model was trained on a BraTS'21 pretrained model

ADC/DWI weight	Dice ($\uparrow$)	Lesion-wise F1 ($\uparrow$)	Lesion count difference ($\downarrow$)	Volume difference ($\downarrow$)
AMC I				
0.6 / 0.4	57.00	62.82*	2.19	1111.57
0.5 / 0.5 (Ours)	60.35	68.30	1.78	806.76
0.4 / 0.6	55.53*	59.28***	2.39**	1054.89
0.3 / 0.7	56.47	<u>64.32</u>	<u>2.04</u>	1165.83
0.2 / 0.8	<u>57.56</u>	63.94	2.12	1023.18
0.1 / 0.9	56.03	62.36*	2.06	<u>1000.85</u>
0.0 / 1.0	42.25***	43.41***	5.57***	1998.92 ****
AMC II				
0.6 / 0.4	69.37***	64.52	3.22	2391.14*
0.5 / 0.5 (Ours)	74.12	67.53	2.91	1564.20
0.4 / 0.6	69.72***	64.47	3.31	2484.93**
0.3 / 0.7	69.18***	64.53	3.05	2304.04
0.2 / 0.8	<u>71.13</u>	<u>66.72</u>	2.91	<u>1939.67</u>
0.1 / 0.9	70.24*	64.83	3.17	2124.60
0.0 / 1.0	46.63****	41.38 ****	9.92****	11,832.39****

Swin UNETR, nnFormer) and NAS-based DiNTS necessitate substantial datasets and prolonged training durations, nnU-Net's powerful auto-configuration settings and framework demonstrate relatively good generalization performance without the need for large datasets and training time, which are difficult to obtain in the medical field. It appears that nnU-Net with robust preprocessing or a well-structured framework generates fewer false positives regardless of lesion location, as evidenced by greater difference in lesion-wise scores compared to voxel-wise scores. These results show why there should be studies that automatically optimize configuration elements that researchers should look for when working with limited datasets.

The ISLES'22 results in Table 3 show that the pretrained models based on the BraTS'21 data can help to learn more suitable representations for ischemic stroke lesion segmentation. Generic features are more robust to noise and perturbation because they typically reflect the high-level characteristics of the data. BraTS'21 includes brain tumor lesions, which are more general than ischemic stroke. Glioblastoma is composed of various cell types such as gemistocyte, granular cell, and metaplasia, and has a solid component with a high density of these cells. This high density of the cells could provoke intra-tumor necrosis, restriction of movement of the water molecules, and intracerebral hemorrhage. However, because ischemic stroke is only caused by oxygen depletion by arterial occlusion without diverse abnormal cell components, the manifestation of the ischemic stroke lesion is relatively uniform than the glioblastoma. In the case of the glioblastoma and ischemic stroke lesions in DWI and ADC, which show the degree of the limitation of the movement of the water molecules, they also have a difference. Because the diffusion restriction caused by glioblastoma is due to the high density of the cell, not the cell death, the severity of the diffusion restriction is milder than ischemic stroke. Thus, the pretraining strategy on BraTS'21 could learn the general representation by differentiating the difficult cases such as small lesions, and the lesion having obscure margins. Thus, the diverse manifestations of the lesion of the glioblastoma play a role to avoid the overfitting problem. To be specific, although brain tumors and ischemic stroke lesions on MRIs have distinct image features, the two brain lesions have appreciable imaging findings compared to normal brain parenchyma, which explains why the BraTS'21 data serves as desirable source data [49]. In our experiment, T2 and T2-FLAIR sequences from the BraTS'21 show superior results for transfer learning. This suggests that gliomas are hyperintense on the T2 imaging, which may be similar to hyperintense ischemic stroke lesions on the DWI in the model's point of view. Further, the transfer learning results using masks from the necrosis class show better overall performance, which may be due to the annular-shaped masks from the enhancing tumor and edema classes that are

unlikely to be found in ischemic stroke images (Table 4). However, our results from two external datasets indicate that only transfer learning is always not guaranteed to improve the performance on the target task on all metrics.

Although Kim et al. [16] demonstrated that segmentation accuracy was increased by utilizing both DWI and ADC images, our study shows that applying both knowledge transfer and joint training can lead to better results when inferring only from the DWI image. We found that using both the ADC and DWI images for inference results in worse generalization performance in terms of the scratch training (Table 5). Our analysis showed that the variances of the DWI images between hospitals, as well as variances of the ADCs, adversely affected the generalization performance. Therefore, we applied the joint training strategy, in which the ADC is simply included in the training phase. However, the performance of joint training from scratch actually became worse because the method of simply using ADC together could hinder learning features from the DWI. On the other hand, joint training with representation transferred from brain tumors allowed the network to jointly learn the features from the DWI and ADC, and to maximize the generalization performance using the DWI alone inference. This experiment shows that even if a particular learning strategy alone performs poorly, its effectiveness can be extremely enhanced when suitably combined with other learning strategies such as transfer learning. This joint approach can be practically applied on various multimodal methods [54].

Generalizability is crucial in medicine, because it guarantees that the results of a study can be utilized to guide clinical practice and healthcare decisions for a wider variety of patients. To evaluate the generalizability of our method, our test datasets were sourced from multiple centers with various devices and European and Asian cohorts. ISLES and BraTS data do not include background noise on MRI, but the AMC I and AMC II datasets contain the noise allowing a pretrained model to make false positives. The false positives especially generate more unconnected lesions degrading lesion-wise metrics. In addition, the AMC I and AMC II datasets are respectively different in that the AMC I was labeled by a neurologist considering patients' clinical symptoms; in other words, ambiguous areas were included if they had to be involved to explain patients' neurological deficits at the time of MRI scan. However, the mask of the AMC II was drawn by radiologists in a conventional way, in that patients' symptoms were not considered. We deduced that the reason a result of AMC I is worse than II because of the ambiguous areas.

Our proposed method not only outperformed competing models at the ISLES'22 challenge (Table 6), but also demonstrated extreme robustness to external data, especially compared to the basic nnU-Net. In addition, this study demonstrates the clinical utility of the largest multimodal public

ischemic stroke dataset by evaluating on real-world hospital clinical data for the first time.

This study has a few limitations. First, it was based on a challenge setting, which might be quite different from an actual clinical setting. We need more multi-center trials and prospective studies. Second, our three datasets include few normal patients. Since deep learning models tend to predict false positives, more normal cases are needed to see if the model predicts false positives for normal patients. Lastly, the pretrained model's sub-optimal performance could limit the fine-tuned model's performance [32]. To avoid this issue, we need to extend our pretrained model's capacity to achieve better adaptation.

Conclusion

In this work, we compare two different multimodal approaches for ischemic stroke lesion segmentation using both ADC and DWI images: joint training and multi-channel training. Transfer learning based on the BraTS'21 dataset shows that it can play an important role in improving stroke lesion segmentation performance with general representation in neuroimaging. Based on the pretrained model, this is the first work reporting 3D joint training based on different modalities, the DWI and ADC, in stroke segmentation. The simultaneous use of transfer learning and joint training for inference with the DWI outperforms channel-wise training. To evaluate the model generalizability performance, we used not only ISLES challenge data (ISLES'22) but also two external cohort datasets from the Asan Medical Center (AMC1 and AMC2). The overall results of ISLES'22, AMC1, and AMC2 show that the multimodal results are significantly better than the single-modal results. In addition, transfer learning boosts the effectiveness of joint training. Finally, our proposed ensemble model achieved the top-1 rank on the ISLES'22 challenge, showing robustness to the challenge presented by external datasets.

Author Contribution Hyunsu Jeong: conceptualization, methodology, software, investigation, validation, formal analysis, writing—original draft, writing—review and editing. Hyunseok Lim: software, visualization, writing—review and editing. Chiho Yoon: software, writing—review and editing. Jongjun Won: supervision, writing—review and editing. Grace Yoojin Lee: supervision, writing—review and editing. Ezequiel de la Rosa: formal analysis. Jan S. Kirschke: formal analysis. Bumjoon Kim: data curation, supervision, writing—review and editing. Namkug Kim: data curation, supervision, writing—review and editing. Chulhong Kim: supervision, writing—review and editing, funding acquisition.

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Data Availability Available upon request.

Declarations

Ethics Approval The AMC I and II datasets were collected according to the principles of the Declaration of Helsinki, and the data collection was performed in accordance with current scientific guidelines. The study protocol was approved by the Institutional Review Board (IRB) Committee of AMC, University of Ulsan College of Medicine, Seoul, Republic of Korea. The requirement for informed patient consent was waived by the IRB Committee of AMC.

Consent to Participate Informed consent was obtained from all individual participants included in the study.

Consent for Publication The authors affirm that human research participants provided informed consent for publication of the images in Figs. 1, 2, 3 and 4.

Conflict of Interest The authors declare no competing interests.

References

1. R.L. Sacco, S.E. Kasner, J.P. Broderick, L.R. Caplan, J.J. Connors, A. Culebras, M.S. Elkind, M.G. George, A.D. Hamdan, R.T. Higashida, B.L. Hoh, L.S. Janis, C.S. Kase, D.O. Kleindorfer, J.M. Lee, M.E. Moseley, E.D. Peterson, T.N. Turan, A.L. Valderrama, H.V. Vinters, C.o.C.S. American Heart Association Stroke Council, Anesthesia, R. Council on Cardiovascular, Intervention, C. Council on, N. Stroke, E. Council on, Prevention, D. Council on Peripheral Vascular, P.A. Council on Nutrition, Metabolism, An updated definition of stroke for the 21st century: a statement for healthcare professionals from the American Heart Association/American Stroke Association, *Stroke*, 44 (2013) 2064–2089.
2. J.L. Saver, Time is brain—quantified, *Stroke*, 37 (2006) 263–266.
3. G. Zaharchuk, I.S. El Mogy, N.J. Fischbein, G.W. Albers, Comparison of arterial spin labeling and bolus perfusion-weighted imaging for detecting mismatch in acute stroke, *Stroke*, 43 (2012) 1843–1848.
4. C. Yoon, S. Misra, K.-J. Kim, C. Kim, B.J. Kim, Collaborative multi-modal deep learning and radiomic features for classification of strokes within 6 h, *Expert Systems with Applications*, 228 (2023) 120473.
5. J. Vymazal, A.M. Rulseh, J. Keller, L. Janouskova, Comparison of CT and MR imaging in ischemic stroke, *Insights Imaging*, 3 (2012) 619–627.
6. B.L. Edlow, S. Hurwitz, J.A. Edlow, Diagnosis of DWI-negative acute ischemic stroke: A meta-analysis, *Neurology*, 89 (2017) 256–262.
7. C.Z. Simonsen, M.H. Madsen, M.L. Schmitz, I.K. Mikkelsen, M. Fisher, G. Andersen, Sensitivity of diffusion- and perfusion-weighted imaging for diagnosing acute ischemic stroke is 97.5%, *Stroke*, 46 (2015) 98–101.
8. E.C. Jauch, J.L. Saver, H.P. Adams, Jr., A. Bruno, J.J. Connors, B.M. Demaerschalk, P. Khatri, P.W. McMullan, Jr., A.I. Qureshi, K.

- Rosenfield, P.A. Scott, D.R. Summers, D.Z. Wang, M. Wintermark, H. Yonas, C. American Heart Association Stroke, N. Council on Cardiovascular, D. Council on Peripheral Vascular, C. Council on Clinical, Guidelines for the early management of patients with acute ischemic stroke: a guideline for healthcare professionals from the American Heart Association/American Stroke Association, *Stroke*, 44 (2013) 870–947.
9. A.K. Rana, J.M. Wardlaw, P.A. Armitage, M.E. Bastin, Apparent diffusion coefficient (ADC) measurements may be more reliable and reproducible than lesion volume on diffusion-weighted images from patients with acute ischaemic stroke-implications for study design, *Magn Reson Imaging*, 21 (2003) 617–624.
10. B.T. Bratane, B. Bastan, M. Fisher, J. Bouley, N. Henninger, Ischemic lesion volume determination on diffusion weighted images vs. apparent diffusion coefficient maps, *Brain Res*, 1279 (2009) 182–188.
11. R. Zhang, L. Zhao, W. Lou, J.M. Abrigo, V.C.T. Mok, W.C.W. Chu, D. Wang, L. Shi, Automatic Segmentation of Acute Ischemic Stroke From DWI Using 3-D Fully Convolutional DenseNets, *IEEE Trans Med Imaging*, 37 (2018) 2149–2160.
12. I. Woo, A. Lee, S.C. Jung, H. Lee, N. Kim, S.J. Cho, D. Kim, J. Lee, L. Sunwoo, D.W. Kang, Fully Automatic Segmentation of Acute Ischemic Lesions on Diffusion-Weighted Imaging Using Convolutional Neural Networks: Comparison with Conventional Algorithms, *Korean J Radiol*, 20 (2019) 1275–1284.
13. K.K. Wong, J.S. Cummock, G. Li, R. Ghosh, P. Xu, J.J. Volpi, S.T.C. Wong, Automatic Segmentation in Acute Ischemic Stroke: Prognostic Significance of Topological Stroke Volumes on Stroke Outcome, *Stroke*, 53 (2022) 2896–2905.
14. P. Ashtari, D.M. Sima, L. De Lathauwer, D. Sappey-Mariniere, F. Maes, S. Van Huffel, Factorizer: A scalable interpretable approach to context modeling for medical image segmentation, *Med Image Anal*, 84 (2023) 102706.
15. L. Liu, L. Kurgan, F.X. Wu, J. Wang, Attention convolutional neural network for accurate segmentation and quantification of lesions in ischemic stroke disease, *Med Image Anal*, 65 (2020) 101791.
16. Y.C. Kim, J.E. Lee, I. Yu, H.N. Song, I.Y. Baek, J.K. Seong, H.G. Jeong, B.J. Kim, H.S. Nam, J.W. Chung, O.Y. Bang, G.M. Kim, W.K. Seo, Evaluation of Diffusion Lesion Volume Measurements in Acute Ischemic Stroke Using Encoder-Decoder Convolutional Network, *Stroke*, 50 (2019) 1444–1451.
17. O. Maier, B.H. Menze, J. von der Gablentz, L. Hani, M.P. Heinrich, M. Liebrand, S. Winzeck, A. Basit, P. Bentley, L. Chen, D. Christiaens, F. Dutil, K. Egger, C.L. Feng, B. Glocker, M. Gotz, T. Haeck, H.L. Halme, M. Havaei, K.M. Iftekharuddin, P.M. Jodoin, K. Kamnitsas, E. Kellner, A. Korvenoja, H. Larochelle, C. Ledig, J.H. Lee, F. Maes, Q. Mahmood, K.H. Maier-Hein, R. McKinley, J. Muschelli, C. Pal, L.M. Pei, J.R. Rangarajan, S.M.S. Reza, D. Robben, D. Rueckert, E. Sallii, P. Suetens, C.W. Wang, M. Wilms, J.S. Kirschke, U.M. Kramer, T.F. Munte, P. Schramme, R. Wiest, H. Handels, M. Reyes, ISLES 2015-A public evaluation benchmark for ischemic stroke lesion segmentation from multispectral MRI, *Medical Image Analysis*, 35 (2017) 250–269.
18. K. Kamnitsas, C. Ledig, V.F.J. Newcombe, J.P. Simpson, A.D. Kane, D.K. Menon, D. Rueckert, B. Glocker, Efficient multi-scale 3D CNN with fully connected CRF for accurate brain lesion segmentation, *Med Image Anal*, 36 (2017) 61–78.
19. O. Ronneberger, P. Fischer, T. Brox, U-Net: Convolutional Networks for Biomedical Image Segmentation, *Medical Image Computing and Computer-Assisted Intervention*, Pt Iii, 9351 (2015) 234–241.
20. R. Karthik, U. Gupta, A. Jha, R. Rajalakshmi, R. Menaka, A deep supervised approach for ischemic lesion segmentation from multimodal MRI using Fully Convolutional Network, *Appl Soft Comput*, 84 (2019).
21. A. Olivier, O. Moal, B. Moal, F. Munsch, G. Okubo, I. Sibon, V. Dousset, T. Tourdias, Active learning strategy and hybrid training for infarct segmentation on diffusion MRI with a U-shaped network, *J Med Imaging (Bellingham)*, 6 (2019) 044001.
22. A. Clerigues, S. Valverde, J. Bernal, J. Freixenet, A. Oliver, X. Llado, Acute and sub-acute stroke lesion segmentation from multimodal MRI, *Comput Methods Programs Biomed*, 194 (2020) 105521.
23. G. Huang, Z. Liu, K.Q. Weinberger, Densely Connected Convolutional Networks, 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), (2016) 2261–2269.
24. A. Kumar, N. Upadhyay, P. Ghosal, T. Chowdhury, D. Das, A. Mukherjee, D. Nandi, CSNet: A new DeepNet framework for ischemic stroke lesion segmentation, *Comput Methods Programs Biomed*, 193 (2020) 105524.
25. Y. Xue, F.G. Farhat, O. Boukrina, A.M. Barrett, J.R. Binder, U.W. Roshan, W.W. Graves, A multi-path 2.5 dimensional convolutional neural network system for segmenting stroke lesions in brain MRI images, *Neuroimage Clin*, 25 (2020) 102118.
26. M.R. Hernandez Petzsche, E. de la Rosa, U. Hanning, R. Wiest, W. Valenzuela, M. Reyes, M. Meyer, S.L. Liew, F. Kofler, I. Ezhov, D. Robben, A. Hutton, T. Friedrich, T. Zarth, J. Burkle, T.A. Baran, B. Menze, G. Broocks, L. Meyer, C. Zimmer, T. Boeckh-Behrens, M. Berndt, B. Ikenberg, B. Wiestler, J.S. Kirschke, ISLES 2022: A multi-center magnetic resonance imaging stroke lesion segmentation dataset, *Sci Data*, 9 (2022) 762.
27. J. Schroeter, C. Myers-Colet, D.L. Arnold, T. Arbel, Segmentation-Consistent Probabilistic Lesion Counting, in: K. Ender, M. Bjoern, V. Archana, B. Christian, D. Qi, A. Shadi (Eds.) *Proceedings of The 5th International Conference on Medical Imaging with Deep Learning*, PMLR, *Proceedings of Machine Learning Research*, 2022, pp. 1034–1056.
28. B. Bozsik, E. Toth, I. Polyak, F. Kerekes, N. Szabo, K. Bencsik, P. Klivenyi, Z.T. Kincses, Reproducibility of Lesion Count in Various Subregions on MRI Scans in Multiple Sclerosis, *Front Neurol*, 13 (2022).
29. O. Commowick, F. Cervenansky, R. Ameli, MSSEG Challenge Proceedings: Multiple Sclerosis Lesions Segmentation Challenge Using a Data Management and Processing Infrastructure, *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2016.
30. H. Zhang, J.W. Zhang, C. Li, E.M. Sweeney, P. Spincemaille, T.D. Nguyen, S.A. Gauthier, Y. Wang, ALL-Net: Anatomical information lesion-wise loss function integrated into neural network for multiple sclerosis lesion segmentation, *Neuroimage-Clin*, 32 (2021).
31. F. Isensee, P.F. Jaeger, S.A.A. Kohl, J. Petersen, K.H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, *Nat Methods*, 18 (2021) 203–211.
32. Y.L. Liu, W.H. Cui, Q. Ha, X.L. Xiong, X.Z. Zeng, C.Y. Ye, Knowledge transfer between brain lesion segmentation tasks with increased model capacity, *Comput Med Imag Grap*, 88 (2021).
33. U. Baid, S. Ghodasara, M. Bilello, S. Mohan, E. Calabrese, E. Colak, K. Farahani, J. Kalpathy-Cramer, F.C. Kitamura, S. Pati, L.M. Prevedello, J.D. Rudie, C. Sako, R.T. Shinohara, T. Bergquist, R. Chai, J.A. Eddy, J. Elliott, W.C. Reade, T. Schaffter, T. Yu, J. Zheng, B. Annotators, C. Davatzikos, J.T. Mongan, C.P. Hess, S. Cha, J.E. Villanueva-Meyer, J.B. Freymann, J.S. Kirby, B. Wiestler, P. Crivellaro, R. R. Colen, A. Kotrotsou, D. Marcus, M. Milchenko, A. Nazeri, H.M. Fathallah-Shaykh, R. Wiest, A. Jakab, M.-A. Weber, A. Mahajan, B.H. Menze, A.E. Flanders, S. Bakas, The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification, *ArXiv*, abs/2107.02314 (2021).
34. A.-R. Lee, I. Woo, D.-W. Kang, S.C. Jung, H. Lee, N. Kim, Fully automated segmentation on brain ischemic and white matter

- hyperintensities lesions using semantic segmentation networks with squeeze-and-excitation blocks in MRI, *Informatics in Medicine Unlocked*, 21 (2020) 100440.
35. M.D. Schirmer, A.K. Giese, P. Fotiadis, M.R. Etherton, L. Cloonan, A. Viswanathan, S.M. Greenberg, O. Wu, N.S. Rost, Spatial Signature of White Matter Hyperintensities in Stroke Patients, *Front Neurol*, 10 (2019) 208.
 36. C. Rorden, M. Brett, Stereotaxic display of brain lesions, *Behav Neurol*, 12 (2000) 191–200.
 37. M. Antonelli, A. Reinke, S. Bakas, K. Farahani, A. Kopp-Schneider, B.A. Landman, ... & M.J. Cardoso, The medical segmentation decathlon. *Nature communications*, 13(1), 4128, (2022).
 38. V. M. Campello, P. Gkontra, C. Izquierdo, C. Martin-Isla, A. Sojoudi, P. M. Full, ... & K. Lekadir, Multi-centre, multi-vendor and multi-disease cardiac segmentation: the M&Ms challenge. *IEEE Transactions on Medical Imaging*, 40(12), 3543–3554, (2021).
 39. J. Wasserthal, H. C. Breit, M. T. Meyer, M. Pradella, D. Hinck, A. W. Sauter, ... & M. Segeroth, Totalsegmentator: Robust segmentation of 104 anatomic structures in ct images. *Radiology: Artificial Intelligence*, 5(5), (2023).
 40. N. Tomita, S. Jiang, M. E. Maeder, S. Hassanpour, Automatic post-stroke lesion segmentation on MR images using 3D residual convolutional neural network, *NeuroImage: clinical*, 27, (2020)
 41. Z. Huang, H. Wang, Z. Deng, J. Ye, Y. Su, H. Sun, J. He, Y. Gu, L. Gu, S. Zhang, Y. Qiao, STU-Net: Scalable and Transferable Medical Image Segmentation Models Empowered by Large-Scale Supervised Pre-training, *arXiv preprint arXiv:2304.06716*, (2023)
 42. Y. He, D. Yang, H. Roth, C. Zhao, D. Xu, DiNTS: Differentiable Neural Network Topology Search for 3D Medical Image Segmentation, *Proceedings of the IEEE international conference on computer vision*, 2021, pp. 5837–5846.
 43. A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H.R. Roth, D. Xu, UNETR: Transformers for 3D Medical Image Segmentation, *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp.574–584
 44. A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H.R. Roth, D. Xu, Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images, *International MICCAI Brainlesion Workshop*, 2021
 45. Y. Gao, M. Zhou, D.N. Metaxas, UTRNet: A Hybrid Transformer Architecture for Medical Image Segmentation, *Medical Image Computing and Computer-Assisted Intervention*, 2021
 46. H.Y. Zhou, J. Guo, Y. Zhang, X. Han, L. Yu, L. Wang, Y. Yu, nnFormer: volumetric medical image segmentation via a 3D transformer, *IEEE Transactions on Image Processing*, 2023
 47. L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, A.L. Yuille, Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs, *IEEE transactions on pattern analysis and machine intelligence*, 40 (2017) 834–848.
 48. M.M.R. Siddique, D. Yang, Y. He, D. Xu, A. Myronenko, Automated ischemic stroke lesion segmentation from 3D MRI, *arXiv preprint arXiv:2209.09546*, (2022).
 49. A. Sasagawa, T. Mikami, Y. Kimura, Y. Akiyama, S. Sugita, T. Hasegawa, M. Wanibuchi, N. Mikuni, Stroke Mimics and Chameleons from the Radiological Viewpoint of Glioma Diagnosis, *Neurol Med Chir (Tokyo)*, 61 (2021) 134–143.
 50. L.M. Ballestar, V. Vilaplana, MRI Brain Tumor Segmentation and Uncertainty Estimation Using 3D-UNet Architectures, *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries*, 2021, 376–390.
 51. Y. Shi, C. Micklisch, E. Mushtaq, S. Avestimehr, Y. Yan, X. Zhang, An ensemble approach to automatic brain tumor segmentation, In *International MICCAI Brainlesion Workshop*, 2021, 138–148
 52. S. T. Henry, A. Carre, M. Lerousseau, T. Estienne, C. Robert, N. Paragios, E. Deutsch, Brain Tumor Segmentation with Self-ensembled, Deeply-Supervised 3D U-Net Neural Networks: A BraTS 2020 Challenge Solution, *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries*, 2021, 327–339.
 53. T. Wiltgen, J. McGinnis, S. Schlaeger, C. Voon, A. Berthele, D. Bischl, L. Grundl, N. Will, M. Metz, D. Schinz, D. Sepp, P. Prucker, B. Schmitz-Koep, C. Zimmer, B. Menze, D. Rueckert, B. Hemmer, J. Kirschke, M. Muhlau, B. Wiestler, LST-AI: a Deep Learning Ensemble for Accurate MS Lesion Segmentation, *medRxiv*, 2023
 54. S. Misra, C. Yoon, K. Kim, R. Managuli, R.G. Barr, J. Baek, and C. Kim, Deep learning-based Multimodal Fusion Network for Segmentation and Classification of Breast Cancers using B-mode and Elastography Ultrasound Images, *Bioengineering and Translational Medicine*, 2023

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